

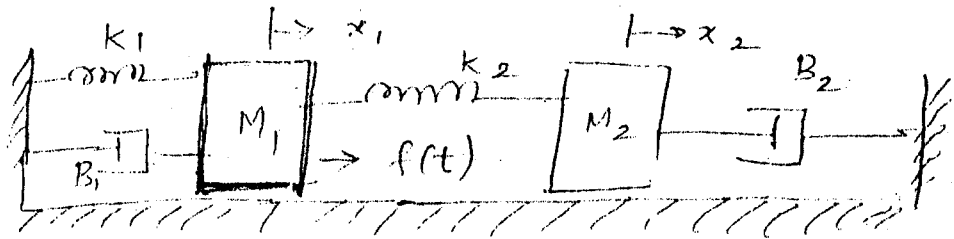
B. Tech Degree VI Semester Examination, June 2006**ME 605 CONTROL SYSTEM ENGINEERING**

(Prior to 2002 Admissions)

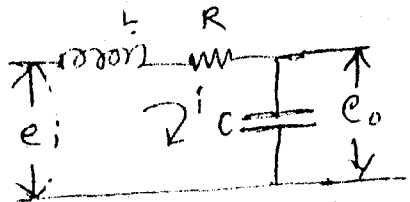
Time : 3 Hours

Maximum Marks : 100

- I. (a) Obtain the Laplace transform of (i) Exponential function (ii) Ramp function. (5)
 (b) Define transfer function of a system. Solve the following differential equation by Laplace transform method :
 $\ddot{x} + 3\dot{x} + 6x = 0, x(0) = 0, \dot{x}(0) = 3.$ (7)
 (c) Distinguish between open loop and closed loop control systems. Give one example for each. (8)
- OR**
- II. (a) Draw a neat sketch of the step response of a second order system and mark the important time domain specifications. (5)
 (b) A unity feedback system has an open loop transfer function $G(s) = \frac{10}{s(s+2)}$. Find rise time, percentage overshoot, peak time and settling time for a step input of 12 units. (10)
 (c) What are static error constants. (5)
- III. (a) For the vibration damper system shown in figure, obtain the force voltage electrical analog. (10)

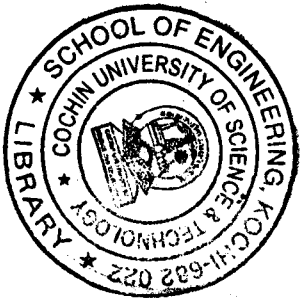


- (b) Obtain the state space representation of the following electrical system. Assume zero initial conditions : (10)

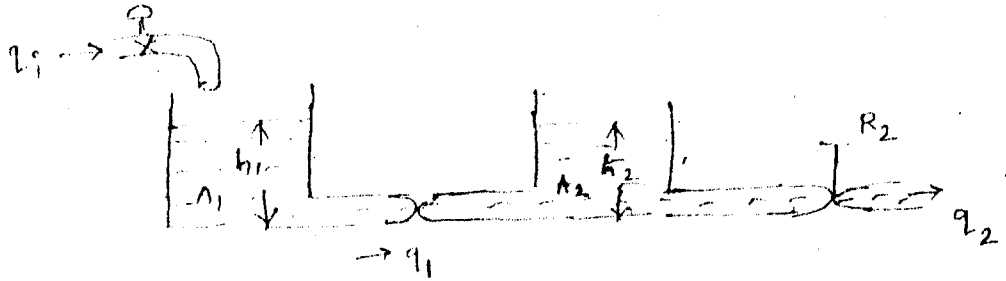
**OR**

- IV. (a) What are the drawbacks of transfer function method of representation of a system? How these drawbacks are eliminated in state variable approach? (8)

(Turn Over)



- (b) For the two tank system shown below, obtain the state equation. (12)



- V. (a) Discuss the stability of a system. (4)
 (b) What are the important frequency domain specifications? (6)
 (c) Determine the range of K for stability of a unity feedback system whose open loop

transfer function is $G(s) = \frac{K}{s(s+1)(s+2)}$. (10)

OR

- VI. (a) What is Routh stability criterion? (6)
 (b) Sketch the Bode plot for the following system and hence determine the gain margin and phase margin of the system. Comment on stability of the system :

$$G(s) = \frac{10}{s(1+0.5s)(1+0.1s)}$$
 (14)

- VII. (a) What are the advantages of digital controllers over analog controllers? (4)
 (b) Find the response $x(k)$ of the following system :

$$x(k+2) - 3x(k+1) + 2x(k) = u(k)$$

where $x(k) = 0$ for $k \leq 0$

$$u(0) = 1$$

$$u(k) = 1 \text{ for } k < 0, k > 0. (8)$$

- (c) Obtain the z-transform of

$$(i) \quad x(n) = \alpha^{-n} u(-n-1)$$

$$(ii) \quad x(n) = n \alpha^n u(-n). (8)$$

OR

- VIII. (a) Discuss sampling theorem. (4)
 (b) Explain the method of determining stability by Jury's test. (6)
 (c) Find the signal corresponding to the z-transform

$$(i) \quad x(z) = \frac{z^{-3}}{2 - 3z^{-1} + z^{-2}}$$

$$(ii) \quad x(z) = \frac{1}{(1 + 0.2z^{-1})(1 - 0.2z^{-1})^2} (10)$$

- IX. (a) Briefly explain integral and derivative control. What are the effects of these controllers on system performance. (10)
 (b) Write notes on :
 (i) Tacho generators (ii) D.C & A.C servomotors. (10)

OR

- X. (a) List the basic control actions in automatic controllers. (8)
 (b) Write notes on :
 (i) Pneumatic controllers (ii) Step per motors
 (iii) Amplidyne. (12)
